RuO_2/TiO_2(110), 4.1 and 7 nm films in 0.1 M H_2SO_4: CTR fits for the CUS OH coverage

26 rod scans (16 of the 4.1 nm film, 10 of the 7 nm film), 2234 points, kinematic model ctrfit; fits of 5 Nov 2026 (files in analysis/2026-11-05_oh_fits/). x_OH = OH share of the oxygen on the CUS Ru, theta (CUS coverage) fixed at 1, OH 2.07 Å above its Ru, H_2O height fitted (2.57 to 2.62 Å).

Results

- **Films.** 4.1 nm fully strained: eps_perp 1.44 to 1.51 %, interface 3.11 to 3.18 Å. 7 nm largely relaxed along [001]: eps_perp 0.50 to 0.64 %, and its (2 0) rod shows no film fringes (the film is out of registry along [001]). Both mean thicknesses ended on their fit bounds, so they compensate something; the fringe periods give about 12.8 and 21.7 trilayers.
- **Only the (0 ±1) rods carry OH information** ($\Delta\chi^2$ 120 to 415 between x_OH 0.3 and 0.7, against 5 or less for (±1 0), (−2 0), (0 2), (2 0)), and they were measured at L 2.5 to 3.4, next to the L = 3 Bragg peak, where x_OH changes |F| by only 7 to 10 % (Fig. 2).
- **The absolute OH coverage is not determined.** On the 13 (0 ±1) scans the best x_OH follows the assumed bridging-O height (Fig. 1: about 0 to 0.45 in A, 0.45 to 0.65 in B, 0.75 to 1 in C) at nearly equal χ^2 (14.5, 14.5, 14.9).
- **Film comparison at the same potential, in all three assumptions:** at −0.1 V and 0 V (shaded) the 4.1 nm film has a higher x_OH than the 7 nm film by about 0.3 to 0.5, i.e. its CUS oxygen sits about 0.2 Å lower on average; at 0.35 and 0.7 V the films agree within error; at OCV the order reverses. Equally possible: the same species sits lower on the more strained film. Not yet a firm result: reduced χ^2 is 14.5, and the film thicknesses are on bounds.
- **Data processing.** The 4.1 nm (0 2) rods have a compressed range at every potential (slope −0.4 of data/model against intensity, Fig. 3; the (0 ±1) rods are near 0), and rod levels differ 3 to 7× within one film and potential. Overall misfit: reduced χ^2 46.5 on all rods, 18.1 on the well-behaved ones, 14.5 on the OH rods.

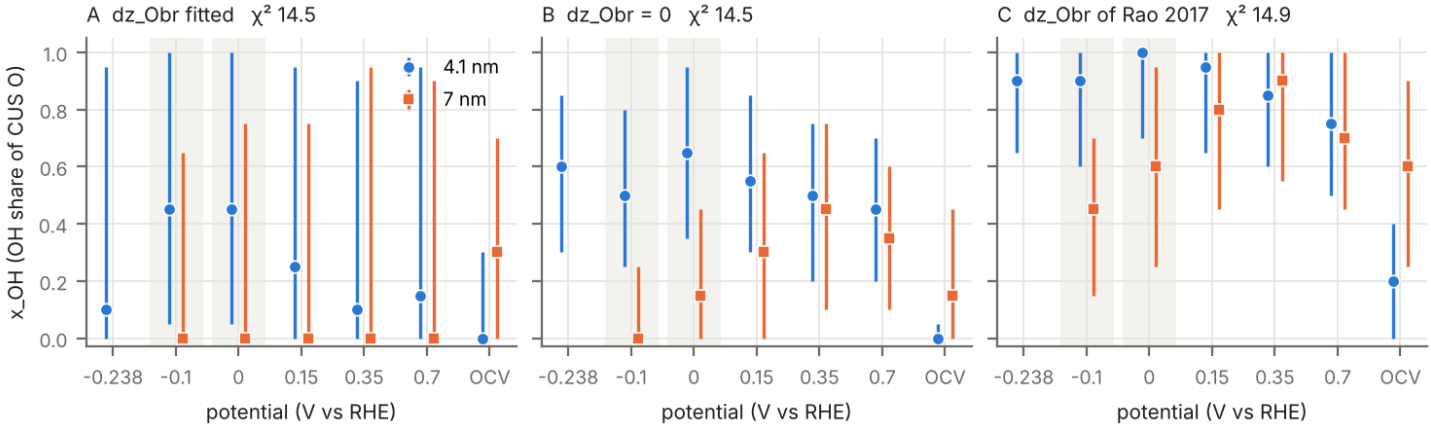


Fig. 1 x_OH with its allowed range (profile, $\Delta\chi^2 \le 4 \times$ reduced χ^2) per film and potential, for three assumptions on the bridging-O height (A: one value per potential shared by the films; reduced χ^2 in the panel titles). The level moves with the assumption; the 4.1 nm − 7 nm difference at −0.1 and 0 V does not.

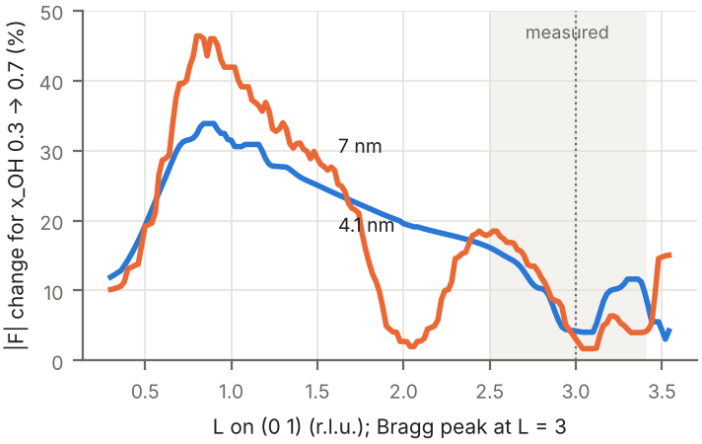


Fig. 2 Where OH changes the (0 1) rod (running median over 0.14 in L): about 35 to 45 % at L 0.7 to 1.1, 7 to 10 % in the measured range (grey).

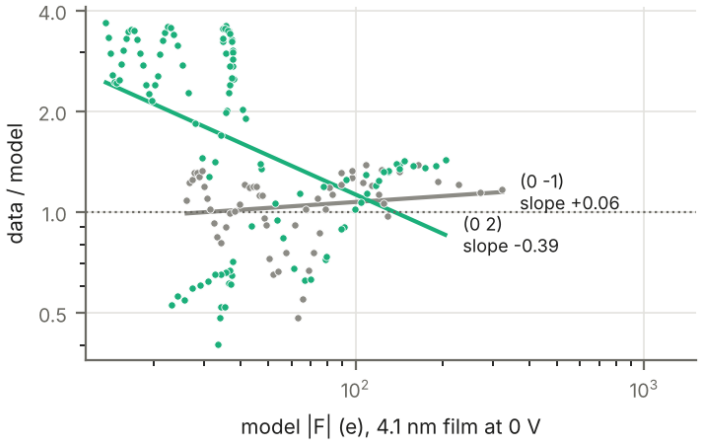


Fig. 3 Data / model against model |F|: the (0 2) rod is compressed (strong points too weak, weak ones too strong), the (0 −1) rod is not. Points within ±0.05 of a Bragg peak left out.

Next steps

- Measure (0 1) at L 0.5 to 2.0 on both films at the same potentials: 3 to 4 times the OH sensitivity (Fig. 2).
- Stationary CVs: (0 −1 1.25) changes 28 to 37 % with x_OH (so far measured only on the 3.3 nm film); the ratio of (0 1 2.75) to (0 1 3.16 / 3.25) cancels the scale and isolates OH (opposite signs, −11 to −14 % against +7 to +10 %).
- Fix the (0 2) processing (attenuator transmission near L 1.86, detector count rate, background at the weak ends) and the rod normalisation; then refit with the thicknesses fixed at their fringe values.